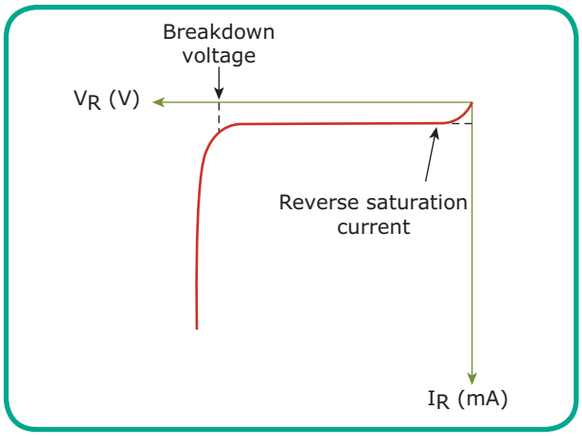


Figure 4.8 Characteristics of PN junction diode (Reverse bias)

The figure 4.8 shows the Characteristics of PN junction diode (Reverse bias)

4.4 TYPE OF RECTIFIERS

There are few type of rectifiers given in Figure 4.9.

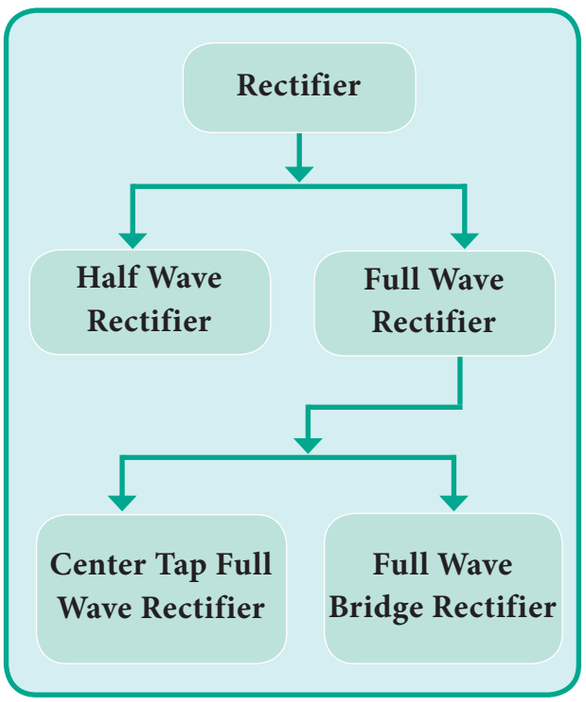


Figure 4.9 Type of Rectifiers

4.4.1 Half-wave Rectifier

This rectifier converts an AC input voltage into pulsating DC voltage for only one half cycle of the applied voltage. The circuit diagram of the half wave rectifier is shown in Figure 4.10(a).

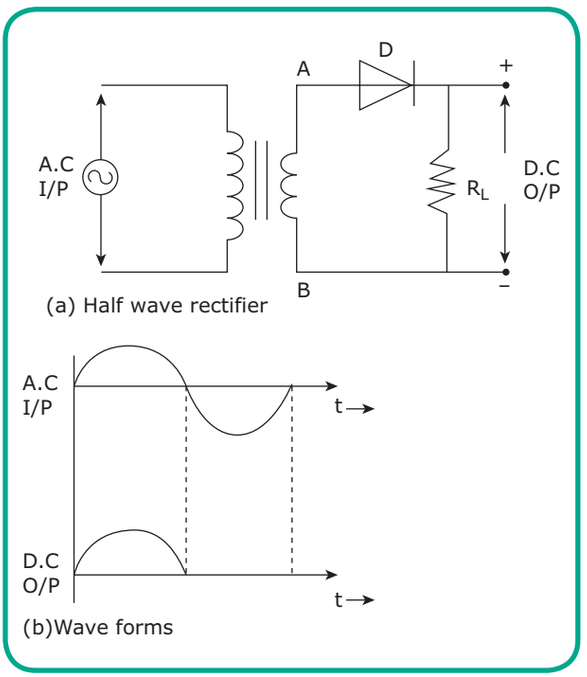


Figure 4.10 Half-wave Rectifier

During the positive half cycle, when the secondary winding of the upper end (A) is positive with respects to the lower end (B), the diode 'D' is under forward bias condition and conducts current.

During the positive half cycles, the input voltage is applied directly to the load resistance R_L, when the forward resistance of the diode is assumed to be zero. The waveforms of the output voltage and output current are same as that of the AC input voltage as shown in Figure 4.10(a).

During the negative half cycle when the lower-end (B) winding of the secondary is positive with respect of the upper-end (A), the diode D is under reverse-bias condition and it does not conduct current. Thus, the load and current across the load resistance is zero and so no power is delivered during the negative half-cycle. The input and output waveforms are shown in Figure 4.10(b).

ELEC_EQUIP_Chapter04.indd 88 11/10/2020 11:38:45 AM